

% %Ä%P%P%P%P%P%P%P%P%P%P%P WebRTC Audio (P2P) %P%P%P%P%P%P%P%P%P%P%

[illegible]

“D

D

****šéːær ˠé-œ ːː**** Audio path = Controller !” Player (direct). Ə

Backend path = signaling messages only.Ⓓ

D

---D

D

4. VPS Environment Variables

D

Backend `.env` / docker-compose:Ð

D

````envD`

STUN SERVERS=stun:stun.l.google.com:19302,stun:stun1.l.google.com:19302

TURN URL=turn:194.233.86.3:34780

TURN\_USERNAME=schoolbell@

TURN PASSWORD=StrongPassword123D

LIVE\_SIGNALING\_TIMEOUT\_MS=3000000

LIVE\_SESSION\_MAX\_MS=7200000D

 $\mathbb{D}$ 

D

```
coturn `/etc/coturn/turnserver.conf` (match credentials):D
```

D

 $\mathbb{D}$ 

listening-port=3478D

```
external-ip=194.233.86.3D
```

user=schoolbell:StrongPassword123D

realm=shikkhasomoy.comĐ

 $\mathbb{D}$ 

D

Firewall: \*\*3478 UDP + TCP\*\* open.Đ

D

---D

D

## ## 5. Authentication

D

API + WebSocket JWT token

D

 $\mathbb{D}$

Authorization: Bearer <JWT>

``

WebSocket connect:

``

wss://api.shikkhasomoy.com/ws/announce?token=<JWT>&role=controller

wss://api.shikkhasomoy.com/ws/announce?token=<JWT>&role=user

``

**\*\*Same user account\*\*** — Controller 2 AE yer ~r 66†ööÂ÷W6W" ¥uB W6R » Ç (same `user\_id`).

---

## ## 6. Complete Live Flow (Step by Step)

### ### Phase A — Session Start (Controller)

| Step | Who | Action |

|-----|-----|-----|

| 1 | Controller | User taps **\*\*Start Live\*\*** |

| 2 | Controller | `POST /api/live/session` `{ role: "controller" }` |

| 3 | Backend | Creates session, returns `session\_id` + `ice\_servers` |

| 4 | Backend | WebSocket ! Player: `{ type: "live\_session", session\_id, ice\_servers }` |

| 5 | Player | Stores `session\_id` + `ice\_servers`, prepares WebRTC |

### ### Phase B — SDP Offer/Answer

| Step | Who | Action |

|-----|-----|-----|

| 6 | Controller | Create `PeerConnection` with `ice\_servers` from step 3 |

| 7 | Controller | Add mic audio track, `createOffer()` |

| 8 | Controller | `POST /api/live/offer` `{ session\_id, role: "controller", offer }` |

| 9 | Backend | WebSocket ! Player: `{ type: "live\_offer", session\_id, offer }` |

| 10 | Player | Create `PeerConnection` with `ice\_servers` (from live\_session or /config) |

| 11 | Player | `setRemoteDescription(offer)` ! `createAnswer()` |

| 12 | Player | `POST /api/live/answer` `{ session\_id, role: "player", answer }` |

| 13 | Backend | WebSocket ! Controller: `{ type: "live\_answer", session\_id, answer }` |

| 14 | Controller | `setRemoteDescription(answer)` |

### ### Phase C — ICE Candidate Exchange (Critical!)

| Step | Who | Action |

|-----|-----|-----|

| 15 | Both | On each local ICE candidate ! `POST /api/live/ice` |

| 16 | Both | On `live\_ice` WS event OR poll ! `GET /api/live/ice?role=...` |

| 17 | Both | `addIceCandidate()` for each peer candidate |

| 18 | Both | PeerConnection state ! **\*\*CONNECTED\*\*** |

| 19 | Player | Remote audio track ! speaker (auto play) |

### ### Phase D — Stop

Step	Who	Action
20	Controller	POST /api/live/end` `{ session_id }`
21	Backend	WebSocket !' both: `{ type: "live_end", session_id, reason }`
22	Both	Close PeerConnection, release mic/audio

## 7. REST API Reference

Method	Path	Role	Purpose
GET	/api/live/config`	Both	STUN/TURN ice_servers
POST	/api/live/session`	Controller	Start session
GET	/api/live/session`	Both	Poll session state
POST	/api/live/offer`	Controller	Send SDP offer
POST	/api/live/answer`	Player	Send SDP answer
POST	/api/live/ice`	Both	Submit ICE candidate
GET	/api/live/ice`	Both	Poll peer ICE candidates
POST	/api/live/end`	Both	End session

### ice\_servers Response Example

```

{
 "ice_servers": [
 {
 "urls": "stun:stun.l.google.com:19302"
 },
 {
 "urls": "stun:stun1.l.google.com:19302"
 },
 {
 "urls": "turn:194.233.86.3:3478",
 "username": "schoolbell",
 "credential": "StrongPassword123"
 }
]
}

```

## 8. WebSocket Messages

```

Controller connects: `role=controller`
Receives: `live_answer`, `live_ice`, `live_end`

Player connects: `role=user` or `role=player`
Receives: `live_session`, `live_offer`, `live_ice`, `live_end`, `new_announcement`

| type | Payload | Action |
|-----|-----|-----|
| `live_session` | session_id, ice_servers | Prepare receiver |
| `live_offer` | session_id, offer | Create answer |

```

| Swagger docs | ``https://api.shikkhasomoy.com/api-docs`` | Ⓓ

Ð  
---Ð

Ð  
## 13. Apps SummaryÐ

Ð  
### Controller AppÐ  
- Starts live sessionÐ  
- Sends offer + ICE candidatesÐ  
- Captures mic, sends audio P2PÐ

Ð  
### Player AppÐ  
- Always connected WebSocket (background service)Ð  
- Receives live\_session + live\_offer automaticallyÐ  
- Sends answer + ICE candidatesÐ  
- Plays remote audio on speakerÐ

Ð  
---Ð

Ð  
\*Smart School Bell — Live WebRTC System Overview\* Ð

\*Backend repository: school-bell-backend\*Ð